

Task 4 – Applications & Shortlisting

Application Funnel Analysis | Data Analyst – Week 2, Phase 2

Objective: Build a decision-ready application funnel that tracks candidates from application through screening, shortlisting, verification, interviews, offers and hiring. The dataset is synthetic and is designed for demonstration, validation and dashboard development.

1. Funnel Definition

The funnel is defined as: Applications → Screening Passed → Shortlisted → Verified Shortlist → Interviewed → Offer → Hired. Each stage is based on an explicit event flag so that the metric can be traced back to individual application records.

2. Key Funnel Results

Stage	Count	Conversion
Applications	1000	100.0%
Screening passed	729	72.9%
Shortlisted	419	41.9%
Verified shortlisted	381	90.9% of shortlisted
Interviewed	270	64.4% of shortlisted
Offers	85	31.5% of interviewed
Hired	59	69.4% of offers

3. Conversion Analysis

The overall application-to-hire conversion is **5.9%**. The largest funnel reduction occurs between applications and screening, where 271 applications do not pass screening. Among shortlisted candidates, 64.4% reach interview. The interview-to-offer conversion is 31.5%, while the offer-to-hire conversion is 69.4%. These stage-level rates identify where operational improvements are most likely to increase successful hires.

4. Source Performance

Source	Applications	Shortlisted	Offers	Hired	Hire Rate
Referral	187	82	20	14	7.5%
Company Website	206	84	20	14	6.8%
Campus	124	55	11	7	5.6%
Job Portal	319	123	25	17	5.3%
LinkedIn	164	75	9	7	4.3%

5. Business Insights

- **Funnel visibility:** The dataset supports an end-to-end view from application to hire, making each KPI explainable from row-level records.
- **Verification control:** 90.9% of shortlisted candidates are marked as verified, supporting a quality-control checkpoint before further action.

- **Source optimisation:** Referral has the highest application-to-hire rate in this synthetic sample at 7.5%, so it is a candidate for further investigation and controlled scaling.
- **Decision use:** A dashboard should show both counts and conversion rates; counts explain volume while rates reveal funnel efficiency.

6. Data Quality & Sanity Checks

Checks applied: unique application IDs, non-null required fields, chronological application timestamps, boolean stage flags, and logical funnel ordering. A later stage is never marked true unless the preceding stage is true. This prevents impossible funnel states and makes the dataset suitable for dashboarding.

7. Recommended Dashboard

Recommended views: (1) KPI cards for applications, shortlisted, offers and hires; (2) funnel chart for stage conversion; (3) source comparison; (4) role/location filters; (5) verification rate; and (6) a detail table for individual applications. Each metric should have a clear definition and source field.

8. Conclusion

The application funnel is complete, traceable and decision-ready. The synthetic dataset demonstrates how student applications can be converted into measurable stages and how companies can use verified shortlisting and conversion metrics to identify bottlenecks and improve hiring outcomes.